task5-da

April 14, 2025

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[2]:
```
sns.set(style="whitegrid")
%matplotlib inline
```

[4]:
```
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/
↪titanic.csv"
df = pd.read_csv(url)
```

[5]:
```
print("Basic Info:\n")
print(df.info())
```

```
Basic Info:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

[6]:
```
print("\n\nSummary Statistics:\n")
print(df.describe())
```

```
Summary Statistics:

       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

[7]:
```
print("\n\nMissing Values:\n")
print(df.isnull().sum())
```

```
Missing Values:

PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

[8]:

```python
print("\n\nValue Counts for 'Sex':")
print(df['Sex'].value_counts())
```

```


Value Counts for 'Sex':
Sex
male      577
female    314
Name: count, dtype: int64
```

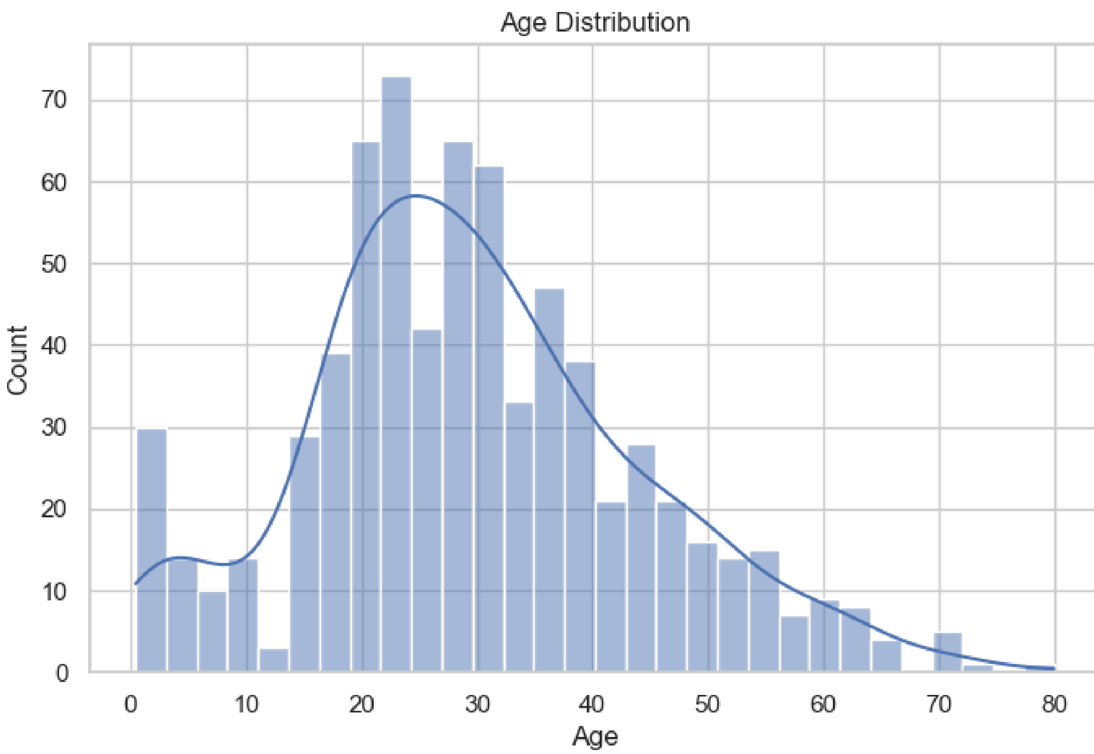
[9]:

```python
print("\n\nValue Counts for 'Pclass':")
print(df['Pclass'].value_counts())
```

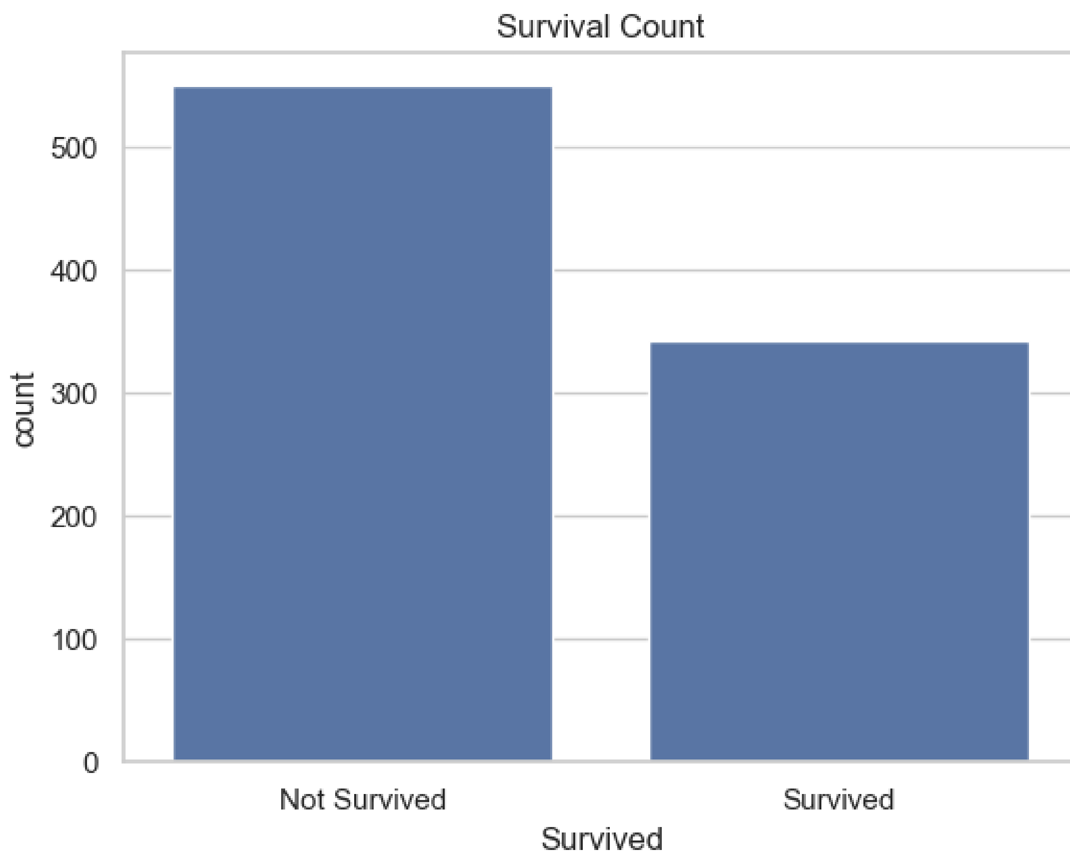
```


Value Counts for 'Pclass':
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

[10]:

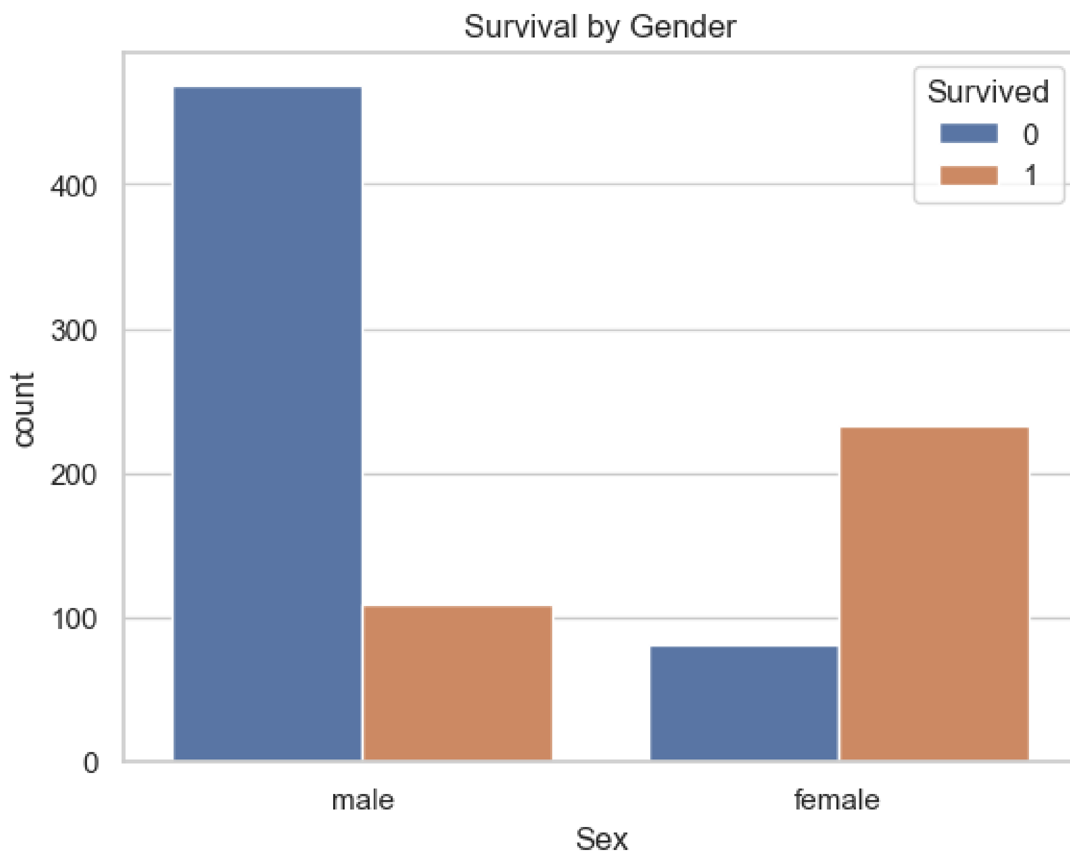
```python
# Histogram for Age
plt.figure(figsize=(8, 5))
sns.histplot(df['Age'].dropna(), bins=30, kde=True)
plt.title('Age Distribution')
plt.show()
```



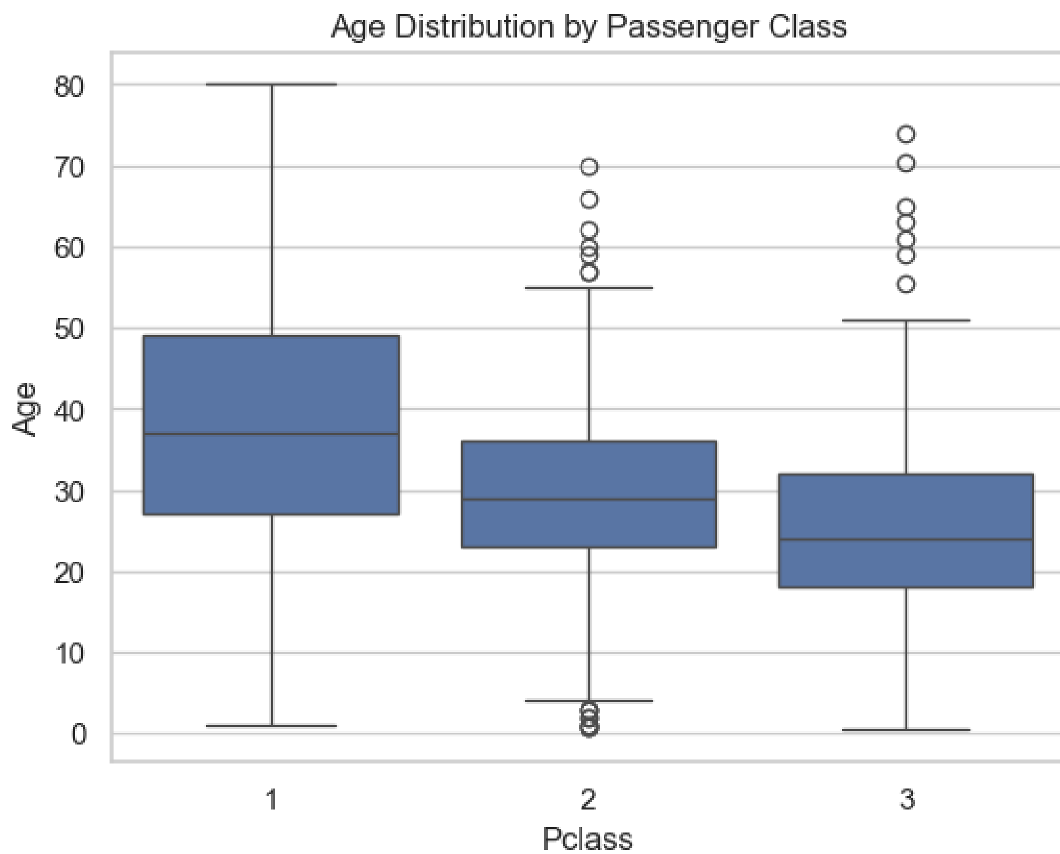
[11]:
```
sns.countplot(x='Survived', data=df)
plt.title('Survival Count')
plt.xticks([0, 1], ['Not Survived', 'Survived'])
plt.show()
```



[12]:
```
# Countplot by Gender
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.show()
```



[13]:
```
# Boxplot of Age vs Pclass
sns.boxplot(x='Pclass', y='Age', data=df)
plt.title('Age Distribution by Passenger Class')
plt.show()
```

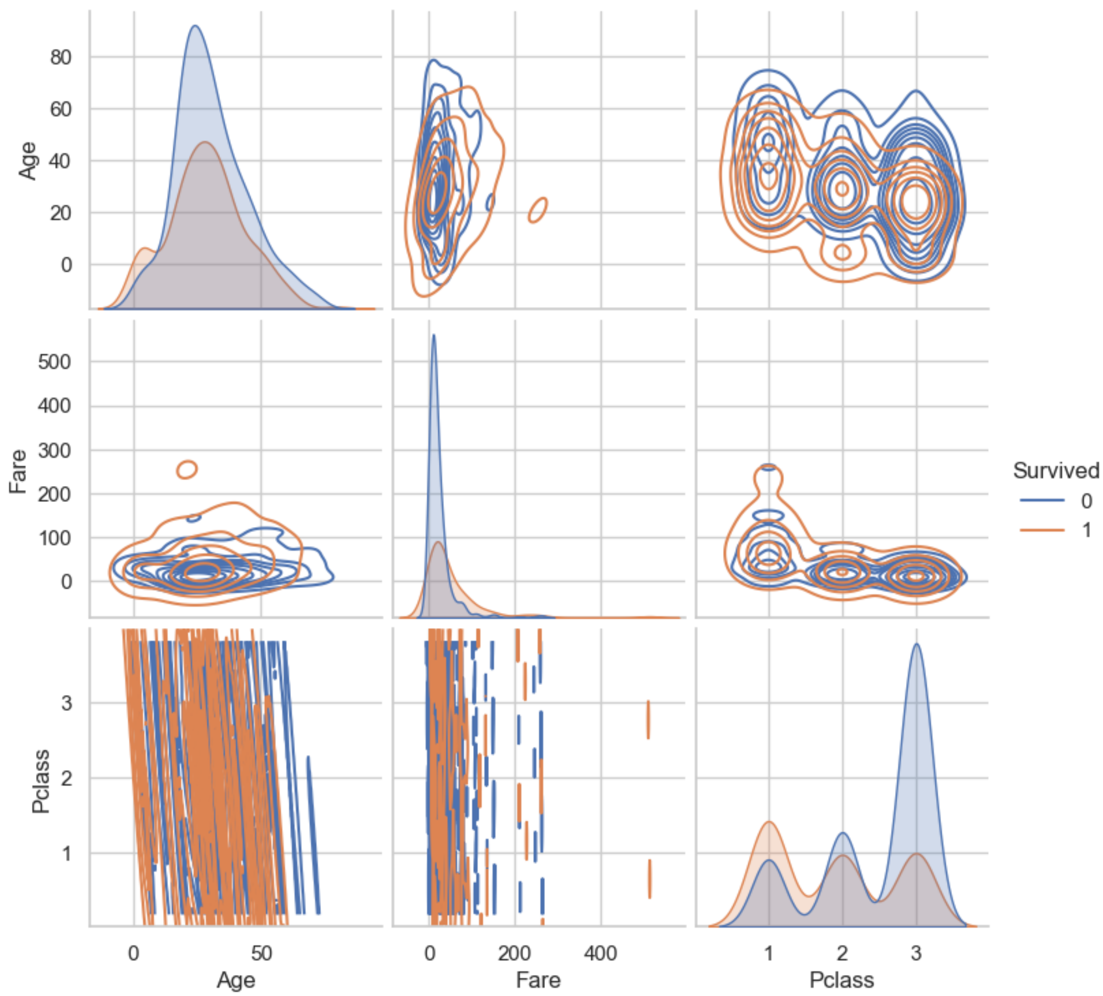


[15]:
```python
df.corr(numeric_only=True)
```

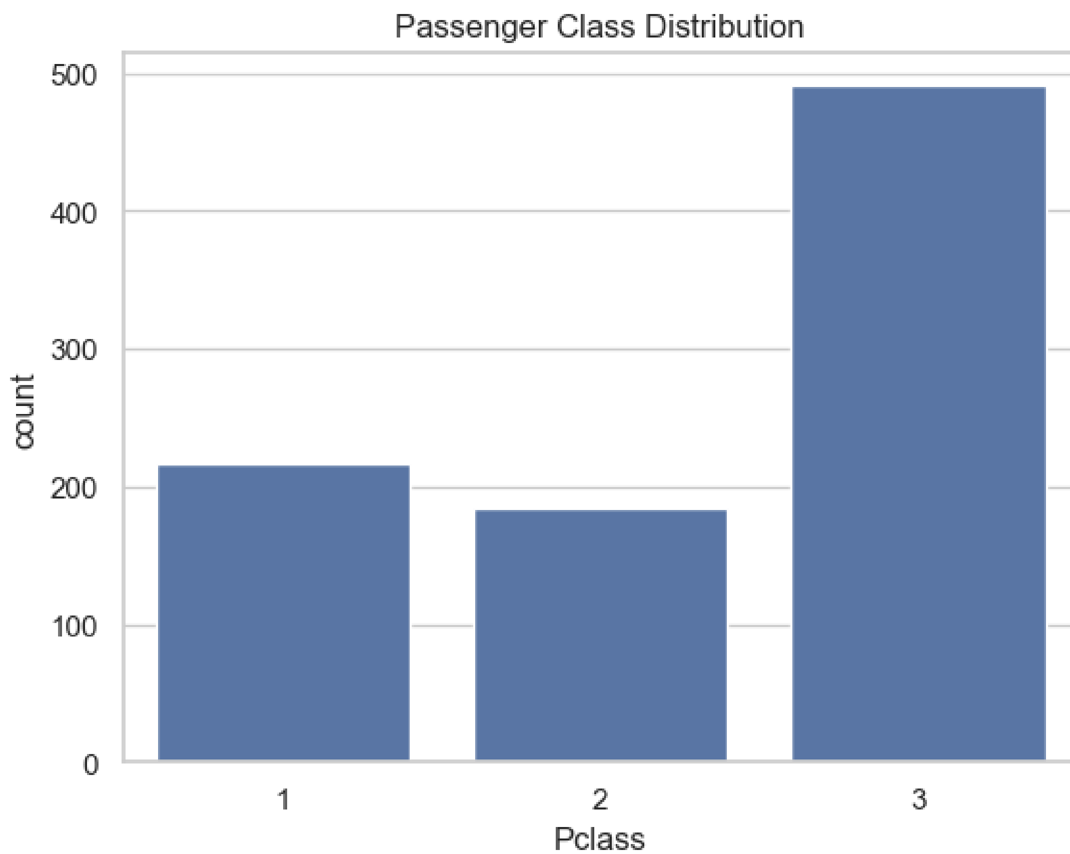
[15]:
```
             PassengerId  Survived    Pclass       Age     SibSp     Parch  \
PassengerId     1.000000 -0.005007 -0.035144  0.036847 -0.057527 -0.001652
Survived       -0.005007  1.000000 -0.338481 -0.077221 -0.035322  0.081629
Pclass         -0.035144 -0.338481  1.000000 -0.369226  0.083081  0.018443
Age             0.036847 -0.077221 -0.369226  1.000000 -0.308247 -0.189119
SibSp          -0.057527 -0.035322  0.083081 -0.308247  1.000000  0.414838
Parch          -0.001652  0.081629  0.018443 -0.189119  0.414838  1.000000
Fare            0.012658  0.257307 -0.549500  0.096067  0.159651  0.216225

                 Fare
PassengerId  0.012658
Survived     0.257307
Pclass      -0.549500
Age          0.096067
SibSp        0.159651
Parch        0.216225
Fare         1.000000
```

[16]:
```
sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']].dropna(),␣
↪hue='Survived', kind='kde')
```

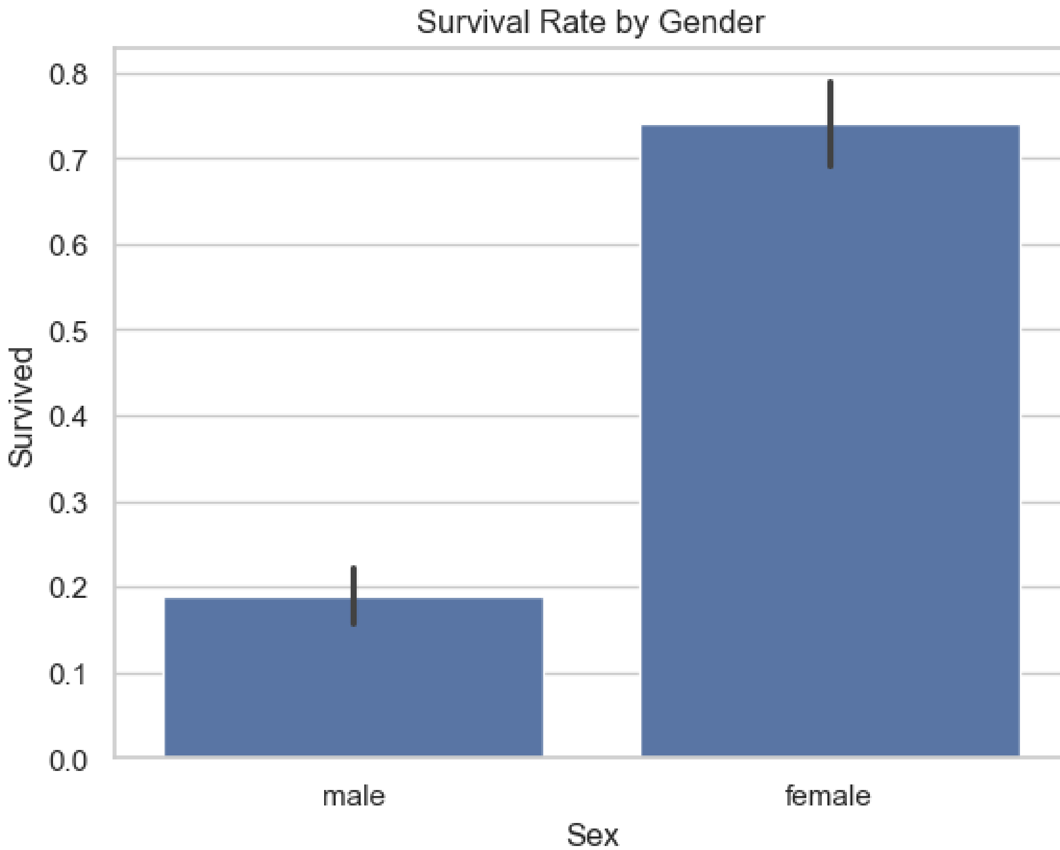
[16]: <seaborn.axisgrid.PairGrid at 0x19fbb7581a0>



[17]:
```
sns.countplot(x='Pclass', data=df)
plt.title('Passenger Class Distribution')
plt.show()
```



[18]:
```
sns.barplot(x='Sex', y='Survived', data=df)
plt.title('Survival Rate by Gender')
plt.show()
```



[19]:
```
print("\nObservations:")
print("""
- Most passengers were in 3rd class and male.
- Females had a higher survival rate than males.
- Younger passengers had slightly higher chances of survival.
- Higher fare and 1st class were positively associated with survival.
""")
```

```
Observations:

- Most passengers were in 3rd class and male.
- Females had a higher survival rate than males.
- Younger passengers had slightly higher chances of survival.
- Higher fare and 1st class were positively associated with survival.
```

[ ]: